

A TWO-CERTIFICATE TRACE-ENERGY DEDUCTION FOR SIMPLE ZEROS OF THE RIEMANN ZETA FUNCTION

YUHAN SHI

ABSTRACT. Let $N(T, 2T)$ count the nontrivial zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in that interval. We refine the finite-block deduction used in recent stability improvements of the rank-trace method. The pressure term in a local gap certificate can be absorbed by a supporting plane obtained from two certified local gap inequalities and the finite-dimensional trace-energy envelope. Applied to the certified seven- and nine-point inequalities on the common window of `trmdy/zeta-simple-zeros-673137`, the refinement gives the research-draft candidate

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6733169771424713 \dots > 0.673316977.$$

No new large numerical search is needed: the only new input is a finite-dimensional inequality proved here. The analytic interface and the machine-certified local inequality are imported at their explicitly stated proof boundary.

1. STATEMENT AND PROOF BOUNDARY

Put

$$S = N_0^s(T, 2T), \quad N = N(T, 2T).$$

We use the arbitrary-window Gabor and stability interface developed in [1, 2] and instantiated with a perturbed admissible window in [4]. In the notation of those sources, it gives

$$(1) \quad S \geq H_{\text{cert}} N + \mathcal{D}(M) - o(N), \quad H_{\text{cert}} = \frac{3362285207}{5000000000},$$

where M is the Gram matrix of the retained simple-zero atoms and

$$\mathcal{D}(G) = \text{tr } \Psi(G), \quad \Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases}$$

For every fixed $R_0 > 0$, the entries of a fixed-size principal Gram block satisfy, uniformly when all its normalized separations are at most R_0 ,

$$(2) \quad G_{ij} = k_v(y_i - y_j) + o(1),$$

where k_v is the normalized overlap kernel of the chosen window. We also use convex spectral pinching and the shifted consecutive-block decomposition of [2, 4].

The finite inputs are two certified inequalities from [4]. Let $w = k_v^2$. For $q \in \{6, 8\}$, let $g_1, \dots, g_q \geq 0$, put $y_0 = 0$ and $y_j = g_1 + \dots + g_j$, and define

$$(\epsilon_q, p_q) = \begin{cases} (891/200000, 1/2736), & q = 6, \\ (15211/2500000, 1/2500), & q = 8. \end{cases}$$

There are nonnegative rational numbers $a_{ij}^{(q)}$ such that

$$(3) \quad p_q \sum_{r=1}^q g_r + \sum_{0 \leq i < j \leq q} a_{ij}^{(q)} w(y_j - y_i) \geq \epsilon_q,$$

$$(4) \quad \sum_{i=0}^{q-r} a_{i,i+r}^{(q)} = 2 \quad (1 \leq r \leq q).$$

The exact 21 and 36 weights and their exhaustive interval certificates are contained in the repository at commit `1610b97b7895ff34982260f8dcdf04a0f7b82cf7`. Our result is conditional only in the sense that (1), (2), and the machine certificate (3) are imported rather than reproved here.

Theorem 1.1. *Subject to the imported interfaces and certificate just specified,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq B,$$

where, with

$$\begin{aligned} A_7 &= \frac{189783}{200000}, & A_9 &= \frac{3209521}{2500000}, \\ R &= 2\sqrt{\frac{349837789}{273750000}} - 1 + \frac{3209521}{547500000}, \\ u &= \frac{R - A_7}{A_9 - A_7}, \\ B &= \frac{219H_{\text{cert}} - (1 - u)213 \cdot 6/2736 - u211 \cdot 8/2500}{219 - R}, \end{aligned}$$

one has

$$B = 0.6733169771424713134 \dots > \frac{673316977}{10^9}.$$

The preceding public assembly of the nine-point certificate gives $0.6733127422722459981 \dots$ [4]; thus the gain in proportion is approximately 4.2349×10^{-6} . All such August 2026 follow-up values are very recent and should be regarded as research-draft candidates pending independent mathematical review.

2. THE SCALED PRESSURE LEMMA

For $m \geq 3$, define

$$(5) \quad \Phi_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ 2\sqrt{\frac{m-1}{m}}E - 1 + \frac{E}{m}, & E \geq m/(m-1). \end{cases}$$

The function is increasing and concave. Indeed, it is linear on the first branch, has negative second derivative on the second branch, and the two one-sided derivatives agree and equal 1 at $E = m/(m-1)$.

Lemma 2.1 (Scaled finite-dimensional envelope). *Let G be an $m \times m$ positive-semidefinite matrix with unit diagonal. If its eigenvalues are $\lambda_1, \dots, \lambda_m$, put*

$$E = \sum_{i=1}^m (\lambda_i - 1)^2, \quad D = \text{tr } \Psi(G).$$

Let $A \geq m/(m-1)$ satisfy $\Phi_m(A) < 2$, set

$$R = \Phi_m(A), \quad \eta = \frac{R}{A},$$

and let $P \geq 0$. If $E + P \geq A$, then

$$(6) \quad D + \eta P \geq R.$$

Proof. Write $x_i = \lambda_i - 1$. Then $x_i \geq -1$, $\sum_i x_i = 0$, and $E = \sum_i x_i^2$. Let

$$L = \{i : x_i > 1\}, \quad q = |L|, \quad X = \sum_{i \in L} x_i, \quad Q = \sum_{i \in L} x_i^2.$$

If $q > 0$, then $q < m$, because $\sum_i x_i = 0$. Since the quadratic branch x_i^2 is replaced by $2x_i - 1$ precisely on L ,

$$(7) \quad D = E + 2X - q - Q.$$

If $q \geq 2$, Cauchy–Schwarz on the $m - q$ complementary coordinates gives $E - Q \geq X^2/(m - q)$. Since $X > q$,

$$(8) \quad D \geq 2X - q + \frac{X^2}{m - q} > q + \frac{q^2}{m - q} = \frac{qm}{m - q} > 2 > R.$$

Suppose next that $q = 1$, and write $x_i = r > 1$ for the unique large coordinate. Cauchy–Schwarz on the remaining $m - 1$ coordinates gives

$$E \geq r^2 + \frac{r^2}{m - 1}, \quad r \leq \sqrt{\frac{m - 1}{m}} E.$$

In particular, the right endpoint in the latter inequality is greater than 1. By (7),

$$D = E + 2r - 1 - r^2.$$

The function $2r - r^2$ decreases for $r \geq 1$, and therefore

$$(9) \quad D \geq \Phi_m(E).$$

If $q = 0$, then simply $D = E$; moreover $q = 0$ necessarily holds when $E < m/(m - 1)$.

The concavity of Φ_m and $\Phi_m(0) = 0$ imply the chord inequality

$$(10) \quad \Phi_m(E) \geq \frac{E}{A} \Phi_m(A) = \eta E \quad (0 \leq E \leq A).$$

Also $R \leq A$, since on the second branch

$$A - \Phi_m(A) = \left(\sqrt{\frac{m - 1}{m}} A - 1 \right)^2 \geq 0.$$

If $E < A$, then (8), (9), the $q = 0$ case, and (10) give either $D > R$ or $D \geq \eta E$. In the latter case $D + \eta P \geq \eta(E + P) \geq R$. If $E \geq A$, the same case analysis gives $D \geq R$: this is immediate when $q \geq 2$; when $q = 1$ it follows from (9) and monotonicity of Φ_m ; and when $q = 0$ it follows from $D = E \geq A \geq R$. This proves (6). $\square$

The next form retains two independent local constraints before paying their global pressure costs.

Lemma 2.2 (Two-certificate supporting plane). *Let G , E , and D be as in Lemma 2.1. Suppose that*

$$E + p_7 L_7 \geq A_7, \quad E + p_9 L_9 \geq A_9, \quad L_7 \geq \frac{3}{4} L_9,$$

where $L_7, L_9 \geq 0$, $0 < A_7 < m/(m - 1) < A_9$, and $R = \Phi_m(A_9) < 2$. Put

$$u = \frac{R - A_7}{A_9 - A_7}, \quad \beta = (1 - u)p_7, \quad \text{quad}\gamma = up_9.$$

If $0 \leq u \leq 1$, $A_7/p_7 \geq 3A_9/(4p_9)$, and

$$(11) \quad \frac{3\beta}{4p_9} + \frac{\gamma}{p_9} \leq 1,$$

then

$$(12) \quad D + \beta L_7 + \gamma L_9 \geq R.$$

Proof. The case $D > R$ is immediate. In every remaining case of the proof of Lemma 2.1, $D \geq \Phi_m(E)$. The two hypotheses and $L_7 \geq 3L_9/4$ give

$$\begin{aligned} L_9 &\geq \max\left(\frac{A_9 - E}{p_9}, 0\right), \\ L_7 &\geq \max\left(\frac{A_7 - E}{p_7}, \frac{3}{4} \frac{A_9 - E}{p_9}, 0\right). \end{aligned}$$

It is thus enough to minimize the resulting lower bound as a function of $E \geq 0$.

On $[0, A_7]$ the first branch of (5) applies. The lower bound is piecewise affine. At $E = 0$ the active lower bound for L_7 is $(A_7 - E)/p_7$ by the extra numerical hypothesis; when the active affine branch changes, the slope remains nonnegative by (11) and by

$$\frac{\beta}{p_7} + \frac{\gamma}{p_9} = 1.$$

Its value at $E = 0$ is R . On $[A_7, A_9]$ it is

$$\Phi_m(E) + \left(\frac{3\beta}{4} + \gamma\right) \frac{A_9 - E}{p_9}.$$

More precisely, the first interval starts at the value R and has nonnegative slope, so its value at A_7 is at least R . The latter function is concave, has that same value at A_7 , and equals R at A_9 ; hence it is at least R throughout. For $E \geq A_9$, monotonicity of Φ_m gives $\Phi_m(E) \geq R$. This proves (12). $\square$

Remark 2.3. The slope η is optimal for a conclusion of the form $D + cP \geq \Phi_m(A)$ based only on $E + P \geq A$: at $E = 0$, $P = A$, one must have $cA \geq \Phi_m(A)$. Thus the improvement below is the full gain available from rescaling this single pressure term at a fixed block length.

3. APPLICATION TO THE SEVEN- AND NINE-POINT CERTIFICATES

Let S° denote the number of retained central simple zeros. The endpoint trimming in the imported interface gives

$$(13) \quad S^\circ = S - o(N).$$

Fix $m = 219$ and partition these retained zeros into consecutive m -point blocks. For a block

$$y_1 < \cdots < y_m,$$

let $W_{q,B}$ be the sum of the spans of all consecutive $(q+1)$ -point windows fully contained in the block:

$$W_{q,B} = \sum_{j=1}^{m-q} (y_{j+q} - y_j) \quad (q = 6, 8).$$

Summing (3) over these $m - q$ windows and using (4), every pair term receives total coefficient at most 2. Hence, with

$$E_B = 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i),$$

we have the two simultaneous estimates

$$(14) \quad E_B + p_q W_{q,B} \geq A_q := \epsilon_q(m - q) \quad (q = 6, 8).$$

Moreover, coefficientwise in the individual gaps,

$$(15) \quad W_{6,B} \geq \frac{3}{4} W_{8,B}.$$

Indeed, an interior gap occurs six times in $W_{6,B}$ and eight times in $W_{8,B}$, while at either endpoint the two multiplicities increase together from one to six.

For $m = 219$,

$$A_7 = \frac{189783}{200000} < \frac{219}{218} < A_9 = \frac{3209521}{2500000},$$

and

$$(16) \quad R = \Phi_{219}(A_9) = 2\sqrt{\frac{349837789}{273750000}} - 1 + \frac{3209521}{547500000} = 1.2667878440823898 \dots < 2.$$

Put $u = (R - A_7)/(A_9 - A_7)$, $\beta = (1 - u)/2736$, and $\gamma = u/2500$. Then $0 < u < 1$, and

$$\frac{\beta}{1/2736} + \frac{\gamma}{1/2500} = 1.$$

Also $3\beta/(4p_9) + \gamma/p_9 < 1$: as a function of u its slope is $1 - 3p_7/(4p_9) = 287/912 > 0$, and it equals 1 at $u = 1$, whereas $u < 1$. Finally,

$$\frac{A_7}{p_7} - \frac{3A_9}{4p_9} = \frac{18909069}{100000} > 0.$$

We claim that every full block satisfies

$$(17) \quad \mathcal{D}(G_B) + \beta W_{6,B} + \gamma W_{8,B} \geq R - o(1),$$

uniformly in the block. If $\beta W_{6,B} + \gamma W_{8,B} \geq R$, this is automatic. Otherwise $W_{8,B} < R/\gamma$, since $\gamma > 0$, so $W_{8,B}$ is uniformly bounded. Because every gap in the block occurs in at least one of its nine-point windows,

$$y_m - y_1 \leq W_{8,B};$$

thus the whole block has uniformly bounded span. The compact-uniform asymptotic (2), with fixed m , gives

$$2 \sum_{i < j} |(G_B)_{ij}|^2 = E_B + o(1)$$

uniformly. The diagonal entries are $1 + o(1)$ in the imported interface. Let $\tilde{G}_B$ be the correlation normalization of G_B . It is positive semidefinite with unit diagonal and, because m is fixed, replacing G_B by $\tilde{G}_B$ changes its finitely many entries, its energy, and $\mathcal{D}$ by $o(1)$ uniformly. For $\tilde{G}_B$ the energy is exactly $\text{tr}(\tilde{G}_B - I)^2 = 2 \sum_{i < j} |(\tilde{G}_B)_{ij}|^2$. Lemma 2.2, continuity of Φ_{219} , and (14)–(15) then prove (17).

It remains to average (17). Use the m shifted partitions into consecutive m -point blocks, retaining an endpoint block of at most $2(m - 1)$ points for each shift. Convex spectral pinching gives

$$\mathcal{D}(M) \geq \sum_B \mathcal{D}(G_B)$$

for every shift. A fixed consecutive $(q + 1)$ -point window is fully contained in a block for at most $m - q$ of the m shifts, with equality away from the two finite endpoints. Endpoint omissions only decrease the pressure sums, whereas the number of full blocks, averaged over the shifts, is $S^\circ/m + O_m(1)$. Moreover,

$$(18) \quad \sum_j (y_{j+q} - y_j) \leq q(y_{S^\circ} - y_1) \quad (q = 6, 8),$$

because every gap occurs in at most q consecutive $(q + 1)$ -point windows. The normalized total span is at most $N + o(N)$ by the Riemann–von Mangoldt formula. The blockwise $o(1)$ in (17) is uniform and the number of blocks is $O(S)$. Consequently

$$(19) \quad \mathcal{D}(M) \geq \frac{R}{m} S^\circ - \frac{\beta 6(m - 6) + \gamma 8(m - 8)}{m} N - o(N).$$

Using (13) replaces S° by S at a cost $o(N)$.

Substituting (19) into (1) and rearranging gives

$$(20) \quad \frac{S}{N} \geq \frac{mH_{\text{cert}} - \beta 6(m-6) - \gamma 8(m-8)}{m-R} - o(1).$$

At $m = 219$, this is the constant B in Theorem 1.1.

For completeness, the strict decimal comparison can be checked using rational arithmetic and one squaring. From (16),

$$R > R_0 := \frac{6333939}{5000000},$$

because

$$\frac{349837789}{273750000} - \left(\frac{R_0 + 1 - A_9/219}{2} \right)^2 = \frac{239006467199}{4796100000000000000} > 0.$$

Since

$$\beta 6(m-6) + \gamma 8(m-8) = \frac{19769000}{31814873}R - \frac{389820601}{3181487300},$$

the right-hand side of (20) is increasing in R : its derivative has the sign of

$$\frac{1799014260305932709}{1590743650000000000} > 0.$$

Replacing R by R_0 therefore gives the rational comparison

$$\frac{23320853620214932709}{34635772470125253000} - \frac{673316977}{10^9} = \frac{4570374547679819}{346357724701252530000000000} > 0.$$

This proves the final strict inequality without relying on rounded decimal arithmetic.

4. RELATION TO THE PRECEDING DEDUCTION

The preceding public finite-block argument uses one local certificate at a time. Lemma 2.2 retains the simultaneous consequences of the certified seven- and nine-point inequalities before their pressure terms are paid globally. The parameter u is the barycentric coordinate of R between A_7 and A_9 , so the resulting pressure tax is a weighted combination of two independently certified taxes. No window, local weight, or interval certificate has been changed.

For comparison, the one-certificate implication is

$$E + P \geq A \implies D + P \geq \Phi_m(A).$$

Lemma 2.1 replaces the coefficient of P by $\eta = \Phi_m(A)/A < 1$. In the shifted-block average this decreases the global pressure tax while preserving the same block defect R . No window, local weights, or interval certificate has been changed.

The Schur–Jensen remainder in [5] is not added separately here. Its scalar function g satisfies $\Psi(t) = g(t) + 1$. If $d_j = M_{jj} \leq 1$ and $\mathcal{S}_g = \text{tr } g(M) - \sum_j g(d_j)$, then the exact identity

$$\text{tr } \Psi(M) - \mathcal{S}_g = \sum_j (1 + g(d_j)) = \sum_j (1 - d_j)^2 \geq 0$$

holds. In particular, the two quantities agree for a unit-diagonal Gram matrix. Adding both terms would therefore double count the same stability slack. The earlier work motivated retaining this information, but the present gain comes specifically from a sharper use of the already retained $\text{tr } \Psi$ defect.

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- Email address:* yuhangshi888@gmail.com